

Data Analysis

Task 4: Clustering

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March 12, 2024

Task description

- ▶ Dataset
 - ▶ Pick the dataset you have previously selected for the 1st task.
 - ▶ If, for some reason, the dataset is unsuitable for this task - you can select another (but confirm it with Tomas)
- ▶ DIY clustering:
 - ▶ Implement a clustering algorithm of your choice **from scratch**.
 - ▶ Popular options - k-means, k-means++, ...
 - ▶ Compare your implementation with scikit-learn version and show that it works correctly.

Task description (2)

- ▶ Data preparation
 - ▶ Perform any data preprocessing if necessary - remove outliers, fix missing data, apply feature scaling, ...
 - ▶ Perform feature selection/engineering: various feature combinations, polynomial features, one-hot encoding, etc...
 - ▶ See if any findings from 1st task (EDA) could be applied to help select suitable features.
- ▶ Clustering
 - ▶ Choose at least 2 different clustering algorithms (from scikit-learn).
 - ▶ One should require specifying the cluster number k (i.e. k-means), the other shouldn't (i.e. DBSCAN)
 - ▶ Apply each algorithm with at least 2 different feature sets (4 different clustering models in total).
 - ▶ Choose one algorithm and adjust its hyperparameters - for example, select cluster number k by using elbow method.

Task description (3)

► Evaluation

- Choose one set of results from the previous step.
- Evaluate clustering quality by using silhouette scores/plots, dendrograms (for hierarchical clustering), vizualizations (scatter plots, PCA), other methods...
- Cluster interpretation: check cluster sizes, centroids, see if the meaning of each cluster can be described.

► Result interpretation

- 5 most important conclusions in domain terms
 - Connection between the numbers and charts and your own knowledge
 - Any unexpected, surprising or otherwise interesting facts about the data? Can you prove/disprove any of your initial assumptions based on the results?

Other info

- ▶ Maximum grade for the task: **1.0**
- ▶ Deadline for task submission to Gitlab: **2024-03-22 16:00**
- ▶ Task submission process:
 - ▶ Jupyter notebook + dataset (.csv file) uploaded to Gitlab, according to the instructions presented in lecture 1
 - ▶ Task presented and explained during exercise class (Tuesdays/Fridays, 18:00 - 19:30)